

Calculate AUC without ROC

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- Calculate AUC directly from tested instances without drawing an ROC curve
- Definition:
 - AUC is the probability that a randomly selected positive example will have a higher probability than a randomly selected negative one

Formula

N_p : number of positive instances

N_n : number of negative instances

$y_1, \dots, y_{N_p}$: score for the positive instances

$x_1, \dots, x_{N_n}$: score for the negative instances

$$AUC = \frac{1}{N_p N_n} \sum_{i=1}^{N_p} \sum_{j=1}^{N_n} \delta(y_i > x_j)$$

Example

Instance	Score	True Class
1	1.0	+
2	0.7	+
3	0.6	+
4	0.5	-
5	0.4	-
6	0.0	-

Classifier 1

$$AUC = \frac{1}{3 * 3} (3 + 3 + 3) = \frac{9}{9}$$

Instance	Score	True Class
1	1.0	+
2	0.9	+
3	0.6	-
4	0.5	+
5	0.2	-
6	0.0	-

Classifier 2

$$AUC = \frac{1}{3 * 3} (2 + 3 + 3) = \frac{8}{9}$$